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May-26-0027

CSPEML-611 (iii) CSPC-615/CSPE-711(ii) (Deep Learning)

B.Tech-6th (NEP) AI&ML (Session 2022-26)
B.Tech.-6th NEP (CSE AIML and CSE AIDS)
B.Tech-7th (NEP) CS&E

Time : 3 Hours **Max. Marks : 60**

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

Section A

1. Compare and contrast single layered model and multi layered perceptron model. (12)

2. Explain Back propagation with its algorithm. Explain different types of back propagation networks. (12)

Section B

3. List and explain the various activation functions used in modeling of artificial neuron. Also explain their suitability with respect to applications. (12)

4. Explain briefly about gradient descent algorithm. (12)

Section C

5. Describe Encoder-Decoder sequence to sequence architecture. (12)

6. What is regularization? How does regularization help in reducing overfitting? (12)

Section D

7. What is a Convolutional Neural Network? Explain in detail. Why is Convolution Neural Network (CNN) preferred for image data? (12)

8. Explain LSTM working principle along with equations. (12)

Section-E (Compulsory)

9. a. List the benefits of activation functions.

   b. Define under fitting.

   c. Define Data Augmentation.

   d. Mention the drawbacks of RNN.

   e. List three stages of a convolutional network.

   f. List some classification problems where Data augmentation is used. (6×2=12)